

Faculty of Science, Engineering and Built Environment

SIT194 Introduction to Mathematical Modelling

Deakin University Unit Guide

Trimester 2, 2025

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Welcome

This unit includes fundamental techniques in differential calculus and integral calculus for the analysis of functions that occur in science and engineering problems. Applications studied include approximations to solutions of equations and integrals and formulation of models to solve science and engineering problems.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: David Tay
Campus: Geelong Waurn Ponds
Email: david.tay@deakin.edu.au
Phone: +61 3 522 78564
Other members of the team and how to contact them

There are no other staff involved in teaching this unit.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

- online learning modules
- active lectures and workshops

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- The feedback for this unit has remained positive, with no suggestions for improvement

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Apply the method of foundational calculus used to solve problems in a range of real-world contexts.	GLO1: Discipline-specific knowledge and capabilities
ULO2	Utilise mathematical modelling to analyse, model, calculate and validate the results associated with solving complex real-life problems.	GLO1: Discipline-specific knowledge and capabilities
ULO3	Evaluate a mathematical model and recommend refinements or improvements.	GLO1: Discipline-specific knowledge and capabilities
ULO4	Critically evaluate solution approaches and tools used in solving problems.	GLO4: Critical thinking
ULO5	Communicate results by showing steps of solution approaches and interpreting the outcomes.	GLO2: Communication

Assessing your achievement of the unit learning outcomes

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Problem solving task 1**
Brief description of assessment task	In this assessment task students are asked to individually complete a series of problems related to topics covered during the first third of the trimester. It includes written answers to mathematical problems, including full explanations of how each answer is derived.
	Problem solving strategy, reasoning, and communication of the solution is important in this task. The intention of the assignments is to assist students in understanding the unit contents, and to provide feedback on how to prepare a mathematical solution.
Detail of student output	This is an individual assessment task. Students will be required to solve problems using appropriate procedures as required. To be successful, students are required to prepare a written mathematical solution clearly, using appropriate wording along with mathematical symbols and notation. Clear and logical explanation of how a solution is obtained is as important as the solution itself.
Grading and weighting (% total mark for unit)	17% graded Students will be given a numerical mark.

This task assesses your achievement of these Unit Learning Outcome(s):	In this assessment students will be evaluated for: ULO1: Apply and communicate the method of foundational calculus used to solve problems in a range of real-world contexts. ULO2: Describe and apply mathematical modelling to analyse, model, calculate and validate the results associated with solving complex real-life problems. ULO4: Critically evaluate solution approaches and tools used in solving problems. ULO5: Communicate results by showing steps of solution approaches and interpreting the outcomes.
This task assesses your achievement of these Graduate Learning Outcome(s)	In this assessment, students will provide evidence of achievement: GLO1: properly using the concepts and tools learned for mathematical modelling GLO2: their ability to communicate their workings/ strategies to solve a problem via a written report GLO4: their ability to critically analyse a solution method and justify the strategy used to solve the problem.
How and when you will receive feedback on your work	Students will receive feedback consisting of a mark, and written comments, generally within ten working days after the submission deadline via the unit site. In addition, they receive feedback on solving similar problems during practicals and lectures.
When and how to submit your work	Students must submit their response to assignment questions electronically, as specified in the assignment description, to the drop-box in the unit site before 8pm AEST by Thursday 31 July, Week 4.

Summative assessment task 2

Title	**Problem solving task 2**
Brief description of assessment task	In this assessment task students are asked to individually complete a series of problems related to topics covered during the second third of the trimester. It includes written answers to mathematical problems, including full explanations of how each answer is derived. Problem solving strategy, reasoning, and communication of the solution is important in this task. The intention of the assignments is to assist students in understanding the unit contents, and to provide feedback on how to prepare a mathematical solution.
Detail of student output	This is an individual assessment task. Students will be required to solve problems using appropriate procedures as required. To be successful, students are required to prepare a written mathematical solution clearly, using appropriate wording along with mathematical symbols and notation. Clear and logical explanation of how a solution is obtained is as important as the solution itself.
Grading and weighting (% total mark for unit)	17% graded Students will be given a numerical mark.
This task assesses your achievement of these Unit Learning Outcome(s)	In this assessment students will be evaluated for: ULO1: Apply and communicate the method of foundational calculus used to solve problems in a range of real-world contexts. ULO2: Describe and apply mathematical modelling to analyse, model, calculate and validate the results associated with solving complex real-life problems. ULO4: Critically evaluate solution approaches and tools used in solving problems. ULO5: Communicate results by showing steps of solution approaches and interpreting the outcomes.
This task assesses your achievement of these Graduate Learning Outcome(s)	This assignment is hand-written, contains mathematical problems to be solved (**GLO1, GLO2, GLO4**), and is to be done independently of other students.
How and when you will receive feedback on your work	Students will receive feedback consisting of a mark, and written comments, generally within ten working days after the submission deadline via the unit site. In addition, they receive feedback on solving similar problems during practicals and lectures.

When and how to submit your work	Students must submit their response to assignment questions electronically, as specified in the assignment description, to the drop-box in the unit site before 8pm AEST by Thursday 28 August 2025, Week 7.

Summative assessment task 3

Title	**Problem solving task 3**
Brief description of assessment task	In this assessment task students are asked to individually complete a series of problems related to topics covered during the last third of the trimester. It includes written answers to mathematical problems, including full explanations of how each answer is derived. Problem solving strategy, reasoning, and communication of the solution is important in this task. The intention of the assignments is to assist students in understanding the unit contents, and to provide feedback on how to prepare a mathematical solution.
Detail of student output	This is an individual assessment task. Students will be required to solve problems using appropriate procedures as required. To be successful, students are required to prepare a written mathematical solution clearly, using appropriate wording along with mathematical symbols and notation. Clear and logical explanation of how a solution is obtained is as important as the solution itself.
Grading and weighting (% total mark for unit)	16% graded Students will be given a numerical mark.
This task assesses your achievement of these Unit Learning Outcome(s):	In this assessment students will be evaluated for: ULO1: Apply and communicate the method of foundational calculus used to solve problems in a range of real-world contexts. ULO2: Describe and apply mathematical modelling to analyse, model, calculate and validate the results associated with solving complex real-life problems. ULO3: Evaluate a mathematical model and recommend refinements or improvements. ULO4: Critically evaluate solution approaches and tools used in solving problems. ULO5: Communicate results by showing steps of solution approaches and interpreting the outcomes.
This task assesses your achievement of these Graduate Learning Outcome(s)	In this assessment, students will provide evidence of achievement: GLO1: properly using the concepts and tools learned for mathematical modelling GLO2: their ability to communicate their workings/ strategies to solve a problem via a written report GLO4: their ability to critically analyse a solution method and justify the strategy used to solve the problem.
How and when you will receive feedback on your work	Students will receive feedback consisting of a mark, and written comments, generally within ten working days after the submission deadline via the unit site. In addition, they receive feedback on solving similar problems during practicals and lectures.
When and how to submit your work	Students must submit their response to assignment questions electronically, as specified in the assignment description, to the drop-box in the unit site before 8pm AEST by Thursday 18 September 2025, Week 10.

Summative assessment task End-of-unit assessment or Examination

Title	**End-of-unit assessment - Written**
Brief description of assessment task	A written assessment that requires the solution of mathematical problems covering the lecture material. This is an individual assessment task.

How you demonstrate your achievement	2-hour written assessment. Students are required to write their response to the questions. To be successful, students are required to prepare a written mathematical solution clearly, using appropriate wording along with mathematical symbols and notation. Clear and logical explanation of how a solution is obtained is as important as the solution itself. A written assessment is a form of assessment that asks you to use text to communicate your knowledge, skill or understanding of a topic. It is completed online. At a set release time, you will be given the assessment instructions and will complete the assessment by a set deadline.
Grading and weighting (% total mark for unit)	50% (Students will receive a final mark and grade for SIT194, based on all assessment tasks. Deakin does not inform students of their end-of-unit mark)
This task assesses your achievement of these Unit Learning Outcome(s)	In this assessment students will be evaluated for: ULO1: Apply and communicate the method of foundational calculus used to solve problems in a range of real-world contexts. ULO2: Describe and apply mathematical modelling to analyse, model, calculate and validate the results associated with solving complex real-life problems. ULO3: Evaluate a mathematical model and recommend refinements or improvements. ULO4: Critically evaluate solution approaches and tools used in solving problems. ULO5: Communicate results by showing steps of solution approaches and interpreting the outcomes.
This task assesses your achievement of these Graduate Learning Outcome(s)	In this assessment, students will provide evidence of achieving in: GLO1: through properly using the concepts and tools learned for mathematical modelling GLO2: through their ability to communicate their workings/ strategies to solve a problem via a written report GLO4: through their ability to critically analyse a solution method and justify the strategy used to solve the problem.
How and when you will receive feedback on your work	Where practicable the Unit Chair will provide general feedback on the assessment task via the unit site in a timely manner. Students can also request individual feedback on their end-of-unit assessment within 10 University working days from receiving their mark. Feedback prior to the end-of-unit assessment is provided through the problem-solving tasks. Students should be able to identify where they need to place effort based on the feedback from these problem-solving tasks which cover all the topics that students will encounter in the end-of-unit assessment. In addition, the students receive feedback on solving similar problems during practicals and lectures. In particular, throughout week 11, a revision of all the concepts will be provided, and problems similar to the ones would appear on the assessment paper will be solved and explained in detail.
When and how to submit your work	Students will be required to undertake a timed online assessment during the end-of-unit assessment period. You will have a start time (e.g., 9am) with a specified time limit of 2 hours. It is the responsibility of students to review their end-of-unit assessment timetable when it is released via StudentConnect. Further information including timetable release dates is available at end-of-unit assessments and examinations.

Formative assessment

(for practice and feedback rather than marks and grades)

In addition to the above, you will have a better chance of success in this Unit if you complete these tasks – you will get valuable experience and feedback.

Details	**Formative quiz**

Brief description	This is an **optional** task, however, students are strongly advised to complete it. In this assessment task students are asked to individually complete a series of online multiple-choice questions to identify the prior knowledge areas and skills they are lacking in. Based on the results of the quiz, students will be directed to further resources to overcome the gaps in their knowledge.
This task assesses your achievement of these Unit Learning Outcome(s)	This online quiz does not contribute to any of the ULOs.
When and how to submit your work	Student will do this quiz before the end of the first week of trimester.
How and when you will receive feedback on your work	Online quiz. Students will receive immediate feedback on the completion of the quiz.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 2 hour online lecture per week, 1 x 2 hour practical experience (workshop) per week, weekly meetings
Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including 1 x 2 hour online lecture per week (recordings provided), 1 x 2 hour online practical experience (workshop) per week, weekly meetings
Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.
For campus students, the learning experiences will include lecture time as described, designated activities in the practical workshop sessions, assessment tasks, readings and study time. For online students, they will include online learning activities, discussion, designated activities in the MS Teams/Zoom sessions, assessment tasks, readings, and study time.

To achieve the SIT194 learning outcomes, students should begin by noting the important dates and times.

The timetable for campus students will be made available before the start of trimester. Please contact the Unit Chair (David Tay) if your timetable does not show **2 weekly online lectures and 1 weekly on-campus practical workshop**.

Students are strongly advised to participate in the lectures and attend Zoom/MS Teams sessions. Even though the notes and lecture recordings will be made available in the unit site, time spent listening and taking notes during lectures reduces the time needed to review the material later.

Students should set aside time to read the unit material, review videos, and attempt the problem set exercises and assessment tasks. All this material is available in the unit site.

The lecture material is related to the questions found in the practicals, the exercises, and the assessments. Completed assessment questions aid in revision for the end-of-unit assessment.

For students who have not studied mathematics for several years, any revision of Year 12 Maths, particularly basic algebra, rules for differentiation and rules for integration will be useful. Any problems or worries can normally be eased by contacting the lecturer.
Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.
Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SIT194 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.
Essential learning resources

The online materials which includes notes and recorded videos constitute the essential learning resources for SIT194.
The SIT194 problem sets and recorded lectures all appear on the unit site and supplement the essential material.
Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

The recommended SIT194 text is "(Single Variable) Calculus; Concepts and Contexts", by James Stewart. Although this text should be quite useful, it is not essential for the successful completion of SIT194. The text is now available in a 4th edition, but any edition will suffice.

Suitable reference books for this unit include:

- Apart from the text by Stewart, there are many texts which cover the first eight weeks of this unit. Most of these have the word calculus in the title, and can be found in the Deakin Library under the 515 lectureification code
- Any edition of the text Calculus and Analytic Geometry, by G.B.Thomas and R.L.Finney is a useful reference, as it contains almost all the material in the unit.

Contact **David Tay** if additional references are required.
Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited

copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 2 teaching period begins	Monday 7 July 2025
Census Date	Friday 15 August 2025
Intra-trimester break	Monday 11 August - Sunday 17 August 2025
Trimester 2 teaching period ends	Friday 26 September 2025
Study period (end-of-unit assessment/examination preparation period)	Monday 29 September - Friday 3 October 2025
End-of-unit assessment/examination period	Monday 6 October - Friday 17 October 2025
Inter-trimester break (the period between trimesters)	Monday 20 October - Friday 31 October 2025
Unit results released	Friday 31 October 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Assessment due date
1	7 July 2025	Odd functions and even functions, functions and inverses	
2	14 July 2025	The derivative, implicit differentiation, evaluation of limits, limits at infinity, L'Hopital's Rule	
3	21 July 2025	Trigonometric functions and inverse, logarithms, hyperbolic functions	
4	28 July 2025	Graph sketching, the definite integral, the Fundamental Theorem of Calculus	**Assessment 1: Problem Solving Task 1** due 31 July
5^	4 August 2025	Integration techniques: substitution; integration by parts; standard integrals	
6	18 August 2025	Partial fractions, Differential Equations, First order differential equations: separable and linear.	
7	25 August 2025	Applications of integration: area; volume. Sequences, series, convergence, ratio test.	**Assessment 2: Problem Solving Task 2** due 28 August
8	1 September 2025	Power series, MacLaurin series, Taylor series, applications in function and integral approximations.	
9	8 September 2025	Multivariate functions: domain and range, level curves and contour lines, partial derivatives.	
10	15 September 2025	Multivariate calculus: implicit differentiation, chain rules, linearization and differentials.	**Assessment 3: Problem Solving Task 3** due 18 September
11#	22 September 2025	Revision	
12	29 September 2025		Study period
13-14	6 October 2025		**End-of-unit assessment**

^ Intra-trimester break: **Monday 11 August – Sunday 17 August 2025**
AFL Grand Final Eve public holiday: **Friday 26 September 2025** - University closed